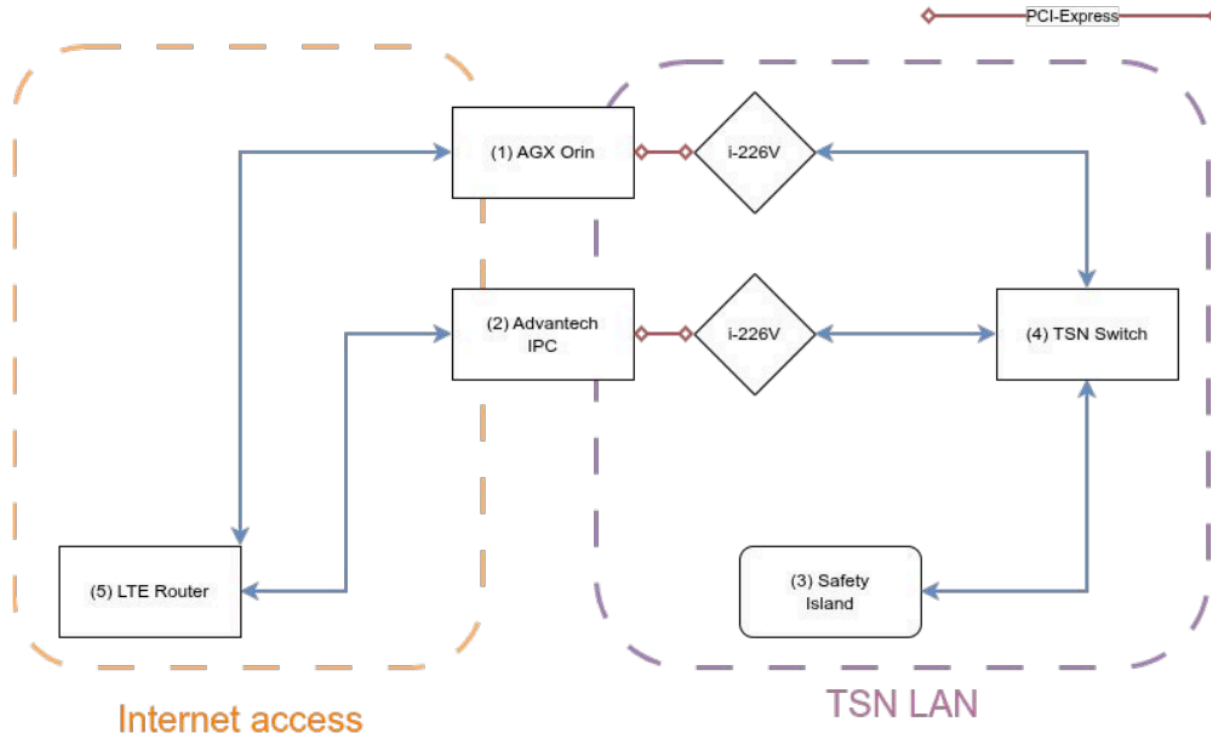


TSN Setup

Time-Sensitive Networking — target architecture

NEWSLab NTU · future work

Target Time-Sensitive Network architecture



Design options towards TSN

All hosts in the TSN must have a network card with TSN features:

- **gPTP** — generalized precision time protocol
- **802.1Qbv** — enhancements for scheduled traffic
- **802.1Qav** — credit-based shaper

A TSN switch is necessary because there are three machines in the network.

The switch cannot be replaced by a normal hub, due to PTP requirements.

TSN support on the machines we have

AGX Orin	I226 NIC attached on the PCIe slot
Advantech	NIC does not support TSN. PCIe is not available and may require slot extension.
NXP safety island	Ethernet T1 socket requires a converter to standard Ethernet TX.
AGX Thor	Supports built-in TSN, but has no PCIe slot. A candidate to replace the AGX Orin.